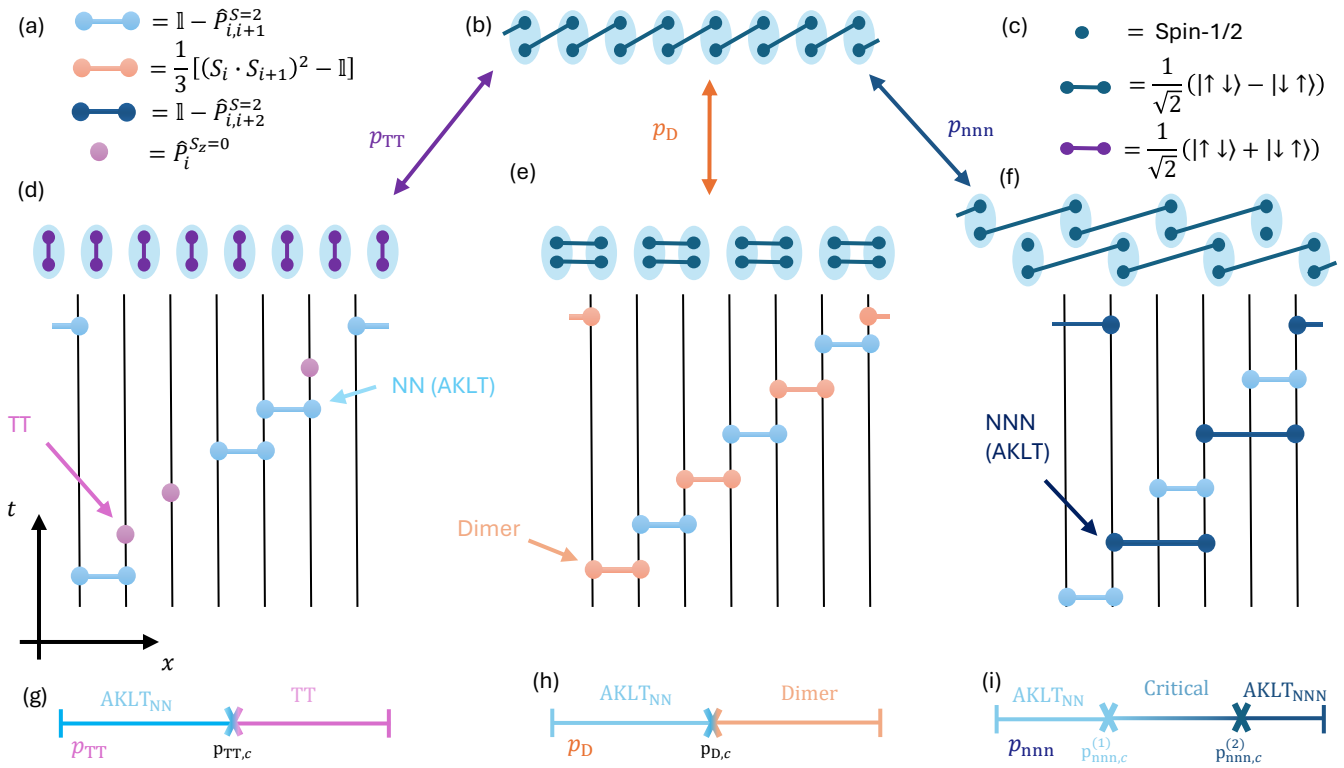

(a)
$= \mathbb{I} - \hat{P}^{S=2}_{i,i+1}$
$= \frac{1}{3}[(S_i \cdot S_{i+1})^2 - \mathbb{I}]$
$= \mathbb{I} - \hat{P}^{S=2}_{i,i+2}$
$= \hat{P}^{S_z=0}_i$
(b)
(c)
$= \text{Spin-1/2}$
$= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$
$= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$
p_{TT}
p_{D}
p_{nnn}
(d)
(e)
(f)
TT
NN (AKLT)
Dimer
NNN
(AKLT)
t
x
(g)
$\mathrm{AKLT_{NN}}$
TT
p_{TT}
$\mathrm{p_{TT,c}}$
(h)
$\mathrm{AKLT_{NN}}$
Dimer
p_{D}
$\mathrm{p_{D,c}}$
(i)
$\mathrm{AKLT_{NN}}$
Critical
$\mathrm{AKLT_{NNN}}$
p_{nnn}
$\mathrm{p^{(1)}_{nnn,c}}$
$\mathrm{p^{(2)}_{nnn,c}}$